# Integration by Substitution

**ANSWER KEY**

## Section 1: Basic u-Substitution

**1** Evaluate $\int 2x(x^2+5)^4\,dx$.

**a.** $\frac{1}{5}(x^2+5)^5+C$ ✓

**b.** $\frac{1}{4}(x^2+5)^4+C$

**c.** $(x^2+5)^5+C$

**d.** $\frac{2}{5}(x^2+5)^5+C$

Let $u=x^2+5 \Rightarrow du=2x\,dx$. $\int u^4\,du=\frac{1}{5}u^5+C=\frac{1}{5}(x^2+5)^5+C$.

**2** Evaluate $\int x^2\sqrt{x^3+1}\,dx$.

**a.** $\frac{2}{9}(x^3+1)^{3/2}+C$ ✓

**b.** $\frac{2}{3}(x^3+1)^{3/2}+C$

**c.** $\frac{1}{3}(x^3+1)^{3/2}+C$

**d.** $\frac{3}{2}(x^3+1)^{1/2}+C$

Let $u=x^3+1 \Rightarrow du=3x^2\,dx$. $\frac{1}{3}\int u^{1/2}\,du=\frac{1}{3}\frac{2}{3}u^{3/2}+C=\frac{2}{9}(x^3+1)^{3/2}+C$.

**3** Evaluate $\int \frac{x}{x^2+9}\,dx$.

**a.** $\frac{1}{2}\ln(x^2+9)+C$ ✓

**b.** $\ln(x^2+9)+C$

**c.** $\frac{1}{3}\arctan(x/3)+C$

**d.** $\frac{1}{2(x^2+9)^2}+C$

Let $u=x^2+9 \Rightarrow du=2x\,dx$. $\frac{1}{2}\int \frac{1}{u}\,du=\frac{1}{2}\ln(x^2+9)+C$.

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## Section 2: Trigonometric and Exponential Substitutions

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**4** Evaluate $\int \sin^3(x)\cos(x)\,dx$.

**a.** $\frac{1}{4}\sin^4(x) + C$ ✓

**b.** $\frac{1}{3}\sin^3(x) + C$

**c.** $-\frac{1}{4}\cos^4(x) + C$

**d.** $\sin^4(x) + C$

Let $u = \sin x \Rightarrow du = \cos x\,dx$. $\int u^3\,du = \frac{1}{4}u^4 + C = \frac{1}{4}\sin^4(x) + C$.

**5** Evaluate $\int \tan(x)\,dx$.

**a.** $\ln|\sec x| + C$ ✓

**b.** $-\ln|\sec x| + C$

**c.** $\sec^2(x) + C$

**d.** $\ln|\sin x| + C$

$\int \frac{\sin x}{\cos x}\,dx$. Let $u = \cos x$, $du = -\sin x\,dx$. $-\int \frac{du}{u} = -\ln|\cos x| + C = \ln|\sec x| + C$.

**6** Evaluate $\int e^{3x-1}\,dx$.

**a.** $\frac{1}{3}e^{3x-1} + C$ ✓

**b.** $3e^{3x-1} + C$

**c.** $e^{3x-1} + C$

**d.** $\frac{1}{3}e^{3x} + C$

Let $u = 3x - 1 \Rightarrow du = 3\,dx$. $\frac{1}{3}\int e^u\,du = \frac{1}{3}e^{3x-1} + C$.

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## Section 3: Definite Integrals with Limits Change

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**7** Evaluate $\int_0^1 x(x^2+1)^3\,dx$.

**a.** 15/8 ✓

**b.** $16/8 = 2$

**c.** 7/8

**d.** 15/4

$u = x^2+1$. When $x=0, u=1$; when $x=1, u=2$. $\frac{1}{2}\int_1^2 u^3\,du = \frac{1}{2}\left[\frac{u^4}{4}\right]_1^2 = \frac{1}{8}[16-1] = \frac{15}{8}$.

**8** Evaluate $\int_0^{\sqrt{\pi}} x\sin(x^2)\,dx$.

**a.** 1 ✓

**b.** 0

**c.** 2

**d.** 1/2

$u = x^2$. $du = 2x\,dx$. Bounds: $x=0 \Rightarrow u=0$: $x=\sqrt{\pi} \Rightarrow u=\pi$.
$\frac{1}{2}\int_0^{\pi}\sin u\,du = \frac{1}{2}[-\cos u]_0^{\pi} = \frac{1}{2}[-(-1)-(-1)] = \frac{1}{2}(2) = 1$.

**9** Evaluate $\int_1^e \frac{\ln x}{x}\,dx$.

**a.** 1/2 ✓

**b.** 1

**c.** 2

**d.** $1/e$

Let $u = \ln x$, $du = dx/x$. Bounds: $u(1)=0, u(e)=1$. $\int_0^1 u\,du = [u^2/2]_0^1 = 1/2$.

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## Section 4: Advanced Linear Substitutions

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**10** Evaluate $\int x\sqrt{x+2}\,dx$.

**a.** $\frac{2}{5}(x+2)^{5/2} - \frac{4}{3}(x+2)^{3/2} + C$ ✓

**b.** $\frac{2}{3}(x+2)^{3/2} + C$

**c.** $\frac{1}{5}(x+2)^{5/2} - \frac{2}{3}(x+2)^{3/2} + C$

**d.** $\frac{2}{5}x^{5/2} + C$

Let $u = x+2 \Rightarrow x = u-2$ and $dx = du$. $\int (u-2)u^{1/2}\,du = \int (u^{3/2} - 2u^{1/2})\,du = \frac{2}{5}u^{5/2} - \frac{4}{3}u^{3/2} + C$.

**11** Evaluate $\int \frac{x}{\sqrt{x+1}}\,dx$.

**a.** $\frac{2}{3}(x+1)^{3/2} - 2\sqrt{x+1} + C$ ✓

**b.** $\frac{1}{3}(x+1)^{3/2} - \sqrt{x+1} + C$

**c.** $2\sqrt{x+1} + C$

**d.** $\frac{2}{3}(x+1)^{3/2} + C$

Let $u = x+1 \Rightarrow x = u-1, dx = du$. $\int (u-1)u^{-1/2}\,du = \int (u^{1/2} - u^{-1/2})\,du = \frac{2}{3}u^{3/2} - 2u^{1/2} + C$.

**12** Evaluate $\int \frac{e^{2x}}{e^x+1}\,dx$.

**a.** $e^x - \ln(e^x+1) + C$ ✓

**b.** $e^x + \ln(e^x+1) + C$

**c.** $\ln(e^x+1) + C$

**d.** $\frac{1}{2}e^{2x} + C$

Let $u = e^x + 1 \Rightarrow e^x = u-1$ and $du = e^x\,dx$.
$\int \frac{u-1}{u}\,du = \int (1 - 1/u)\,du = u - \ln|u| + C = e^x + 1 - \ln(e^x+1) + C = e^x - \ln(e^x+1) + C'$.

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